

SkillSync - AI Career Growth & Upskilling Assistant

Problem Statement

In today's rapidly evolving job market, professionals and students struggle to identify which skills to acquire next to remain relevant. Traditional career guidance systems are static, generic, and fail to provide actionable, data-driven insights. SkillSync aims to solve this by providing an AI-powered personalized career and skill development assistant that analyzes user profiles, identifies skill gaps, tracks real-time job trends, and recommends learning paths, courses, and projects - all tailored to individual goals and market demand.

Objectives

1. Build a web platform that uses AI + data analytics to guide users on skill development.
2. Enable users to connect LinkedIn or upload resumes to extract their current skills.
3. Use AI models to suggest personalized learning paths, project ideas, and resume generation.
4. Fetch live job market trends to recommend high-demand technologies.
5. Provide dashboards for tracking progress and insights.

Technology Stack & Tools

Frontend: React.js, Tailwind CSS, Axios, React Router DOM, Redux/Zustand

Backend: Node.js, Express.js, JWT, OpenAI API, MongoDB

Database: MongoDB Atlas

AI: OpenAI GPT-4 / Gemini APIs for AI recommendations

Tools: VS Code, Postman, Render, GitHub, Docker (optional)

Functional Requirements

- User Authentication: Register/Login using JWT or OAuth.
- Profile Analysis: Extract skills from LinkedIn or resume.
- AI Career Advisor: Chat-based GenAI for personalized guidance.

- Roadmap Generator: Personalized skill learning paths.
- Resume Builder: AI-generated resumes and cover letters.
- Project Recommender: Suggests practical projects.
- Job Market Insights: Display trending roles and technologies.
- Dashboard: Track progress and goals.
- Admin Panel: Manage users and data.

System Architecture

Frontend (React.js)

- > Backend (Node.js + Express)
- > MongoDB (User data, roadmaps)
- > AI Layer (OpenAI/Gemini APIs)
- > External APIs (LinkedIn, Job Market APIs)

Development Steps

Phase 1: Project setup with MERN structure and environment variables.

Phase 2: Implement authentication using JWT.

Phase 3: Build AI integration layer (OpenAI/Gemini API).

Phase 4: Create resume parser and skill extractor.

Phase 5: Integrate job trend APIs and analytics dashboard.

Phase 6: Develop AI Chat Assistant and personalized dashboard.

Phase 7: Implement AI Resume Builder and project recommender.

Phase 8: Add Admin Panel for management (optional).

Phase 9: Deploy using Render/Vercel + MongoDB Atlas.

Future Enhancements

- AI Voice Assistant integration (Whisper + TTS)
- Chrome extension for analyzing LinkedIn jobs

- Mentor-based AI chatbot
- Gamified learning system and badges